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McGee et al.

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(54) **VALBENAZINE SALTS AND POLYMORPHS THEREOF**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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Related U.S. Application Data

(63) Continuation of application No. 16/662,346, filed on Oct. 24, 2019, now abandoned, which is a continuation of application No. 16/293,728, filed on Mar. 6, 2019, now abandoned, which is a continuation of application No. 16/043,059, filed on Jul. 23, 2018, now abandoned, which is a continuation of application No. 15/338,214, filed on Oct. 28, 2016, now Pat. No. 10,065,952.

(60) Provisional application No. 62/249,074, filed on Oct. 30, 2015.

(51) **Int. Cl.**
C07D 471/04 (2006.01)
A61K 31/4375 (2006.01)

(52) **U.S. Cl.**
CPC ***C07D 471/04*** (2013.01); *C07B 2200/13* (2013.01)

(58) **Field of Classification Search**
CPC C07D 471/04; C07B 2200/13
See application file for complete search history.

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(57) **ABSTRACT**

Provided herein are salts of (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester in amorphous and crystalline forms, and processes of preparation, and pharmaceutical compositions thereof. Also provided are methods of their use for treating, preventing, or ameliorating one or more symptoms of neurological disorders and diseases including hyperkinetic movement disorders or diseases.

27 Claims, 18 Drawing Sheets

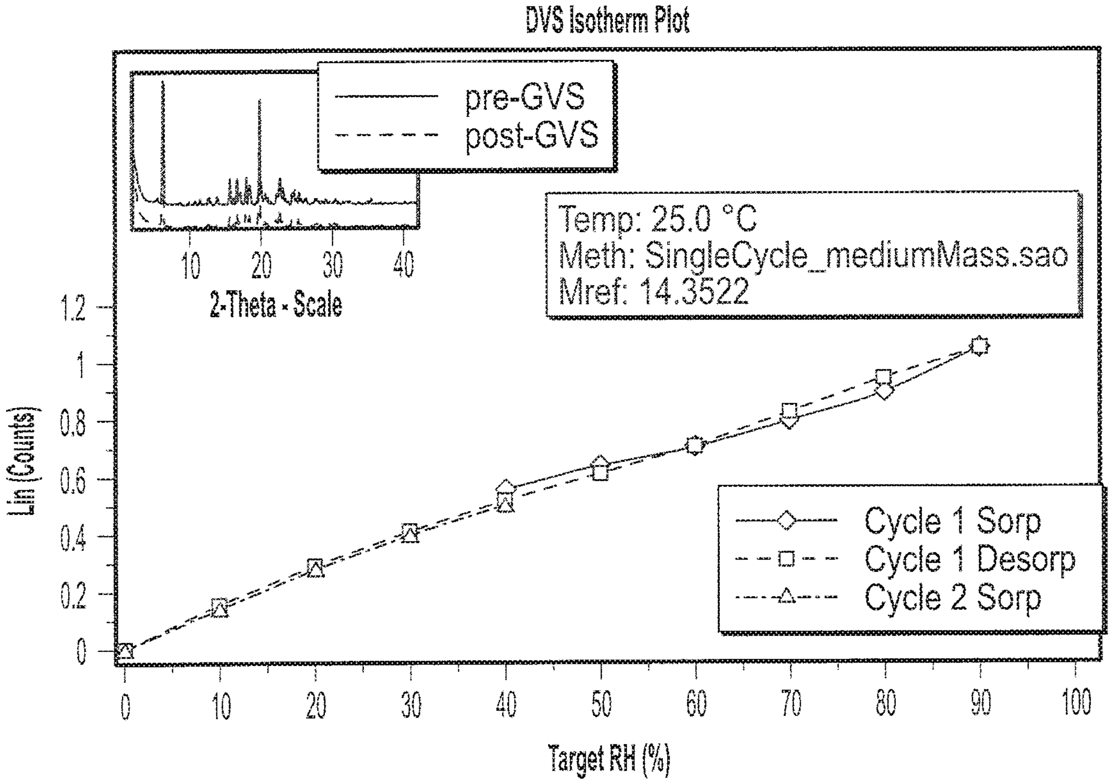

DVS Isotherm Plot
pre-GVS
post-GVS
2-Theta - Scale
10
20
30
40
Temp: 25.0 °C
Meth: SingleCycle_mediumMass.sao
Mref: 14.3522
Lin (Counts)
0
0.2
0.4
0.6
0.8
1
1.2
Cycle 1 Sorp
Cycle 1 Desorp
Cycle 2 Sorp
0
10
20
30
40
50
60
70
80
90
100
Target RH (%)

FIG. 3

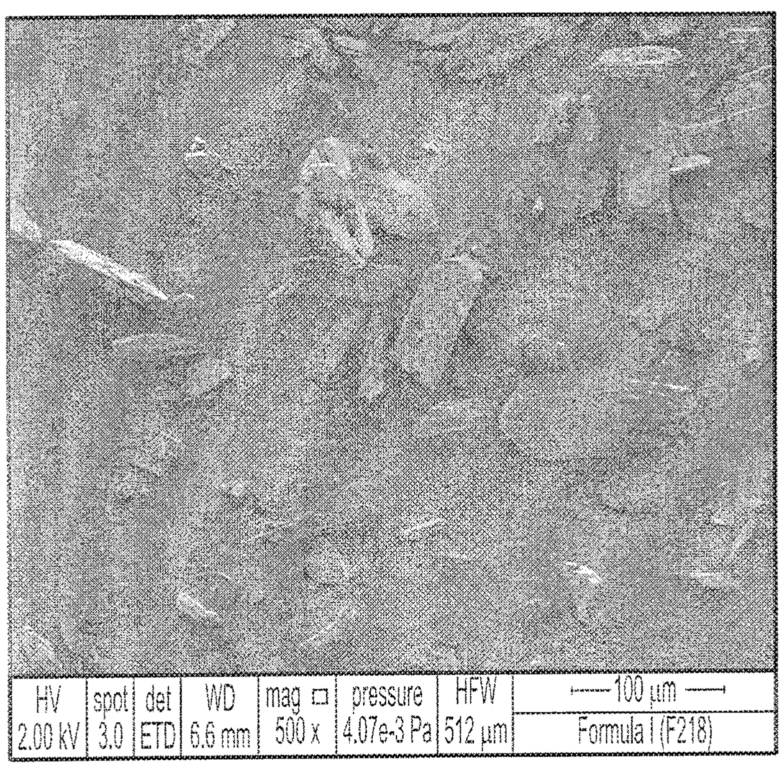

HV 2.00 kV
spot 3.0
det ETD
WD 6.6 mm
mag 500 x
pressure 4.07e-3 Pa
HFW 512 µm
100 µm
Formula I (F218)

FIG. 4A

VALBENAZINE SALTS AND POLYMORPHS THEREOF

CROSS REFERENCE TO RELATED APPLICATION

This application claims the benefit of U.S. application Ser. No. 16/662,346 filed Oct. 24, 2019, which claims the benefit of U.S. application Ser. No. 16/293,728 filed Mar. 6, 2019, which claims the benefit of U.S. application Ser. No. 16/043,059 filed Jul. 23, 2018, which claims the benefit of U.S. application Ser. No. 15/338,214 filed Oct. 28, 2016, now U.S. Pat. No. 10,065,952, which claims the benefit of U.S. Provisional Application No. 62/249,074 filed Oct. 30, 2015; the disclosure of each of which is incorporated herein by reference in its entirety.

FIELD

Provided herein are salts of (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester in amorphous and crystalline forms, processes of preparation thereof, and pharmaceutical compositions thereof. Also provided are methods of their use for treating, preventing, or ameliorating one or more symptoms of neurological disorders and diseases including hyperkinetic movement disorders or diseases.

BACKGROUND

Hyperkinetic disorders are characterized by excessive, abnormal involuntary movement. These neurologic disorders include tremor, dystonia, ballism, tics, akathisia, stereotypies, chorea, myoclonus and athetosis. Though the pathophysiology of these movement disorders is poorly understood, it is thought that dysregulation of neurotransmitters in the basal ganglia plays an important role. (Kenney et. al., *Expert Review Neurotherapeutics,* 2005, 6, 7-17). The chronic use and high dosing of typical neuropletics or centrally acting dopamine receptor blocking antiemetics predispose patients to the onset of tardive syndromes. Tardive dyskinesia, one subtype of the latter syndromes, is characterized by rapid, repetitive, stereotypic, involuntary movements of the face, limbs, or trunk. (Muller, *Expert Opin. Investig. Drugs,* 2015, 24, 737-742).

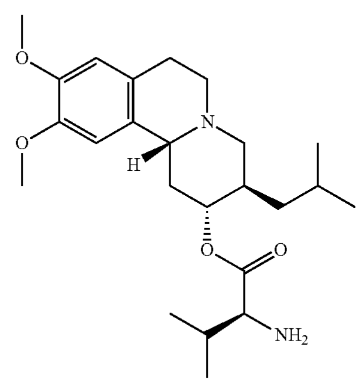
The reversible inhibition of the vesicular monoamine transporter-2 system (VMAT2) by 3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-one, also known as tetrabenazine (TBZ), improves the treatment of various hyperkinetic movement disorders. However, the drawbacks to such treatment are the fluctuating response, the need for frequent intake do to TBZ rapid metabolism, and side effects. Side effects associated with TBZ include sedation, depression, akathisia, and parkinsonism.

TBZ, which contains two chiral centers and is a racemic mix of two stereoisomers, is rapidly and extensively metabolized in vivo to its reduced form, 3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-ol, also known as dihydrotetrabenazine (DHTBZ). DHTBZ is thought to exist as four individual isomers: (±) alpha-DHTBZ and (±) beta-DHTBZ. The 2R, 3R, 11bR or (+) alpha-DHTBZ is believed to be the absolute configuration of the active metabolite. (Kilbourn et al., *Chirality,* 1997, 9, 59-62). Tetrabenazine has orphan drug status in US and is approved in certain European countries. Its use is also allowed for therapy of chorea in patients with Hungtington's disease. However, tetrabenazine is rapidly metabolized and must frequently be administered throughout the day. (Muller, *Expert Opin. Investig. Drugs,* 2015, 24, 737-742). Therefore, there is an unmet need in the art to develop effective therapeutics for treatment of hyperkinetic movement disorders, including tardive dyskinesia.

Valbenazine, (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester, the purified prodrug of the (+)-α-isomer of dihydrotetrabenazine, recently showed a distinctive improvement in the treatment of hyperkinetic movement disorders, including tardive dyskinesia symptoms, with improved pharmacokinetic and tolerability profiles.

SUMMARY OF THE DISCLOSURE

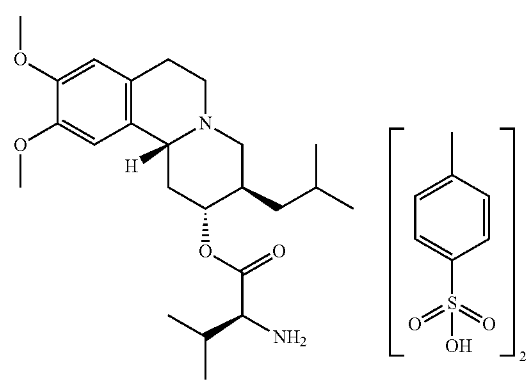
Provided herein are pharmaceutically acceptable salts of (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,1-a]isoquinolin-2-yl ester of Formula:



or an isotopic variant thereof; or solvate thereof.

Provided herein is a crystalline form of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) of Formula I

(I)



or an isotopic variant thereof; or solvate thereof.

Also provided herein are Forms I, II, III, IV, V, and VI of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-aminowise, all technical and scientific terms used herein generally have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs.

The term "subject" refers to an animal, including, but not limited to, a primate (e.g., human), cow, pig, sheep, goat, horse, dog, cat, rabbit, rat, or mouse. The terms "subject" and "patient" are used interchangeably herein in reference, for example, to a mammalian subject, such as a human subject, in one embodiment, a human.

As used herein, "isotopically enriched" refers to an atom having an isotopic composition other than the natural isotopic composition of that atom. "Isotopically enriched" may also refer to a compound containing at least one atom having an isotopic composition other than the natural isotopic composition of that atom.

With regard to the compounds provided herein, when a particular atomic position is designated as having deuterium or "D," it is understood that the abundance of deuterium at that position is substantially greater than the natural abundance of deuterium, which is about 0.015%. A position designated as having deuterium typically has a minimum isotopic enrichment factor of, in particular embodiments, at least 1000 (15% deuterium incorporation), at least 2000 (30% deuterium incorporation), at least 3000 (45% deuterium incorporation), at least 3500 (52.5% deuterium incorporation), at least 4000 (60% deuterium incorporation), at least 4500 (67.5% deuterium incorporation), at least 5000 (75% deuterium incorporation), at least 5500 (82.5% deuterium incorporation), at least 6000 (90% deuterium incorporation), at least 6333.3 (95% deuterium incorporation), at least 6466.7 (97% deuterium incorporation), at least 6600 (99% deuterium incorporation), or at least 6633.3 (99.5% deuterium incorporation) at each designated deuterium position.

The isotopic enrichment of the compounds provided herein can be determined using conventional analytical methods known to one of ordinary skill in the art, including mass spectrometry, nuclear magnetic resonance spectroscopy, and crystallography.

Isotopic enrichment (for example, deuteration) of pharmaceuticals to improve pharmacokinetics ("PK"), pharmacodynamics ("PD"), and toxicity profiles, has been demonstrated previously with some classes of drugs. See, for example, Lijinsky et. al., *Food Cosmet. Toxicol.,* 20: 393 (1982); Lijinsky et. al., *J. Nat. Cancer Inst.,* 69: 1127 (1982); Mangold et. al., *Mutation Res.* 308: 33 (1994); Gordon et. al., *Drug Metab. Dispos.,* 15: 589 (1987); Zello et. al., *Metabolism,* 43: 487 (1994); Gately et. al., *J. Nucl. Med.,* 27: 388 (1986); *Wade D, Chem. Biol. Interact.* 117: 191 (1999).

Isotopic enrichment of a drug can be used, for example, to (1) reduce or eliminate unwanted metabolites, (2) increase the half-life of the parent drug, (3) decrease the number of doses needed to achieve a desired effect, (4) decrease the amount of a dose necessary to achieve a desired effect, (5) increase the formation of active metabolites, if any are formed, and/or (6) decrease the production of deleterious metabolites in specific tissues and/or create a more effective drug and/or a safer drug for combination therapy, whether the combination therapy is intentional or not.

Replacement of an atom for one of its isotopes often will result in a change in the reaction rate of a chemical reaction. This phenomenon is known as the Kinetic Isotope Effect ("KIE"). For example, if a C—H bond is broken during a rate-determining step in a chemical reaction (i.e. the step with the highest transition state energy), substitution of a deuterium for that hydrogen will cause a decrease in the reaction rate and the process will slow down. This phenomenon is known as the Deuterium Kinetic Isotope Effect ("DKIE"). (See, e.g., Foster et al., *Adv. Drug Res.*, vol. 14, pp. 1-36 (1985); Kushner et al., *Can. J. Physiol. Pharmacol.*, vol. 77, pp. 79-88 (1999)).

The magnitude of the DKIE can be expressed as the ratio between the rates of a given reaction in which a C—H bond is broken, and the same reaction where deuterium is substituted for hydrogen. The DKIE can range from about 1 (no isotope effect) to very large numbers, such as 50 or more, meaning that the reaction can be fifty, or more, times slower when deuterium is substituted for hydrogen. High DKIE values may be due in part to a phenomenon known as tunneling, which is a consequence of the uncertainty principle. Tunneling is ascribed to the small mass of a hydrogen atom, and occurs because transition states involving a proton can sometimes form in the absence of the required activation energy. Because deuterium has more mass than hydrogen, it statistically has a much lower probability of undergoing this phenomenon.

Tritium ("T") is a radioactive isotope of hydrogen, used in research, fusion reactors, neutron generators and radiopharmaceuticals. Tritium is a hydrogen atom that has 2 neutrons in the nucleus and has an atomic weight close to 3. It occurs naturally in the environment in very low concentrations, most commonly found as T_2O. Tritium decays slowly (half-life=12.3 years) and emits a low energy beta particle that cannot penetrate the outer layer of human skin. Internal exposure is the main hazard associated with this isotope, yet it must be ingested in large amounts to pose a significant health risk. As compared with deuterium, a lesser amount of tritium must be consumed before it reaches a hazardous level. Substitution of tritium ("T") for hydrogen results in yet a stronger bond than deuterium and gives numerically larger isotope effects. Similarly, substitution of isotopes for other elements, including, but not limited to, ^{13}C or ^{14}C for carbon, ^{33}S, ^{34}S, or ^{36}S for sulfur, ^{15}N for nitrogen, and ^{17}O or ^{18}O for oxygen, may lead to a similar kinetic isotope effect.

For example, the DKIE was used to decrease the hepatotoxicity of halothane by presumably limiting the production of reactive species such as trifluoroacetyl chloride. However, this method may not be applicable to all drug classes. For example, deuterium incorporation can lead to metabolic switching. The concept of metabolic switching asserts that xenogens, when sequestered by Phase I enzymes, may bind transiently and re-bind in a variety of conformations prior to the chemical reaction (e.g., oxidation). This hypothesis is supported by the relatively vast size of binding pockets in many Phase I enzymes and the promiscuous nature of many metabolic reactions. Metabolic switching can potentially lead to different proportions of known metabolites as well as altogether new metabolites. This new metabolic profile may impart more or less toxicity.

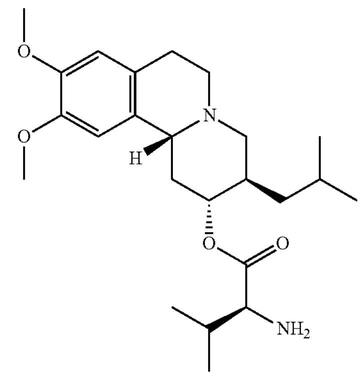
The animal body expresses a variety of enzymes for the purpose of eliminating foreign substances, such as therapeutic agents, from its circulation system. Examples of such enzymes include the cytochrome P450 enzymes ("CYPs"), esterases, proteases, reductases, dehydrogenases, and monoamine oxidases, to react with and convert these foreign substances to more polar intermediates or metabolites for renal excretion. Some of the most common metabolic reactions of pharmaceutical compounds involve the oxidation of a carbon-hydrogen (C—H) bond to either a carbon-oxygen (C—O) or carbon-carbon (C—C) pi-bond. The resultant metabolites may be stable or unstable under physiological conditions, and can have substantially different pharmacokinetic, pharmacodynamic, and acute and long-term toxicity hydroxynaphthoic, salicylic, stearic, cyclohexylsulfamic, quinic, muconic acid and the like acids; or (2) salts formed when an acidic proton present in the parent compound either (a) is replaced by a metal ion, e.g., an alkali metal ion, an alkaline earth ion or an aluminum ion, or alkali metal or alkaline earth metal hydroxides, such as sodium, potassium, calcium, magnesium, aluminum, lithium, zinc, and barium hydroxide, ammonia, or (b) coordinates with an organic base, such as aliphatic, alicyclic, or aromatic organic amines, such as ammonia, methylamine, dimethylamine, diethylamine, picoline, ethanolamine, diethanolamine, triethanolamine, ethylenediamine, lysine, arginine, ornithine, choline, N,N'-dibenzylethylene-diamine, chloroprocaine, diethanolamine, procaine, N-benzylphenethylamine, N-methylglucamine piperazine, tris(hydroxymethyl)-aminomethane, tetramethylammonium hydroxide, and the like.

Pharmaceutically acceptable salts further include, by way of example only and without limitation, sodium, potassium, calcium, magnesium, ammonium, tetraalkylammonium, and the like, and when the compound contains a basic functionality, salts of non-toxic organic or inorganic acids, such as hydrohalides, e.g. hydrochloride and hydrobromide, sulfate, phosphate, sulfamate, nitrate, acetate, trifluoroacetate, trichloroacetate, propionate, hexanoate, cyclopentylpropionate, glycolate, glutarate, pyruvate, lactate, malonate, succinate, sorbate, ascorbate, malate, maleate, fumarate, tartarate, citrate, benzoate, 3-(4-hydroxybenzoyl)benzoate, picrate, cinnamate, mandelate, phthalate, laurate, methanesulfonate (mesylate), ethanesulfonate, 1,2-ethane-disulfonate, 2-hydroxyethanesulfonate, benzenesulfonate (besylate), 4-chlorobenzenesulfonate, 2-naphthalenesulfonate, 4-toluenesulfonate, camphorate, camphorsulfonate, 4-methylbicyclo[2.2.2]-oct-2-ene-1-carboxylate, glucoheptonate, 3-phenylpropionate, trimethylacetate, tert-butylacetate, lauryl sulfate, gluconate, benzoate, glutamate, hydroxynaphthoate, salicylate, stearate, cyclohexylsulfamate, quinate, muconate, and the like.

The term "amino acid" refers to naturally occurring and synthetic a, R, y, or 6 amino acids, and includes but is not limited to, amino acids found in proteins, i.e. glycine, alanine, valine, leucine, isoleucine, methionine, phenylalanine, tryptophan, proline, serine, threonine, cysteine, tyrosine, asparagine, glutamine, aspartate, glutamate, lysine, arginine and histidine. In one embodiment, the amino acid is in the L-configuration. Alternatively, the amino acid can be a derivative of alanyl, valinyl, leucinyl, isoleuccinyl, prolinyl, phenylalaninyl, tryptophanyl, methioninyl, glycinyl, serinyl, threoninyl, cysteinyl, tyrosinyl, asparaginyl, glutaminyl, aspartoyl, glutaroyl, lysinyl, argininyl, histidinyl, β-alanyl, β-valinyl, β-leucinyl, β-isoleuccinyl, β-prolinyl, β-phenylalaninyl, β-tryptophanyl, β-methioninyl, β-glycinyl, β-serinyl, β-threoninyl, β-cysteinyl, β-tyrosinyl, β-asparaginyl, β-glutaminyl, β-aspartoyl, β-glutaroyl, β-lysinyl, β-argininyl, or β-histidinyl.

Solid Forms

In one embodiment, provided herein are pharmaceutically acceptable salts of (S)-2-amino-3-methyl-butyric acid (2R, 3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester or an isotopic variant thereof. (S)-2-amino-3-methyl-butyric acid (2R,3R, 11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,1-a]isoquinolin-2-yl ester has the structure of Formula:

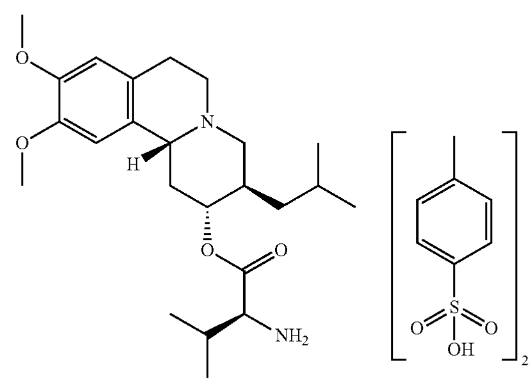


The compound (S)-2-amino-3-methyl-butyric acid (2R, 3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester, also known as valbenazine, can be prepared according to U.S. Pat. Nos. 8,039,627 and 8,357,697, the disclosure of each of which is incorporated herein by reference in its entirety.

Valbenazine Ditosylate

In another embodiment, provided herein is a crystalline form of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3, 4,6,7,11b-hexahydro-1H-pyrido[2,1-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) or an isotopic variant thereof or solvate thereof of Formula I

(I)



The crystalline forms as shown herein (e.g., of Formula I) may be characterized using a number of methods known to a person skilled in the art, including single crystal X-ray diffraction, X-ray powder diffraction (XRPD), microscopy (e.g., scanning electron microscopy (SEM)), thermal analysis (e.g., differential scanning calorimetry (DSC), thermal gravimetric analysis (TGA), and hot-stage microscopy, and spectroscopy (e.g., infrared, Raman, solid-state nuclear magnetic resonance). The particle size and size distribution may be determined by conventional methods, such as laser light scattering technique. The purity of the crystalline forms provided herein may be determined by standard analytical methods, such as thin layer chromatography (TLC), gel electrophoresis, gas chromatography, high performance liquid chromatography (HPLC), and mass spectrometry (MS).

Valbenazine Ditosylate Form I

In yet another embodiment, provided herein is a crystalline form of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,1-a]isoquinolin-2-yl greater than about 1%, no greater than about 2%, no greater than about 3%, no greater than about 4%, or no greater than about 5% water by weight.

In certain embodiments Form I may be characterized by particle analysis. In certain embodiments, a sample of Form I comprises particles having rhomboid crystal morphology. In yet another embodiment, a sample of Form I comprises particles of about 100, about 90, about 80, about 70, about 60, about 50, about 40, about 30, about 20, about 10, about 5 μM in length. In some embodiments, a sample of Form I comprises particles of about 70, about 60, about 40, about 20, about 10 μM in length. In other embodiments, a sample of Form I comprises particles of about 69.39, about 56.22, about 34.72, about 17.84, about 10.29 μM in length.

Valbenazine Ditosylate Form II

In another embodiment, is a crystalline form of (S)-(2R, 3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) or an isotopic variant thereof or solvate thereof; wherein the crystalline form is Form II.

In various embodiments, crystalline Form II of (S)-(2R, 3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) has an X-ray diffraction pattern. In some embodiments, the X-ray diffraction pattern of Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) includes an XRP diffraction peak at two-theta angles of approximately 5.7, 15.3, and 22.5°. In some embodiments, the X-ray powder diffraction pattern of Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) includes an XRP diffraction peak at two-theta angles of approximately 5.7, 15.3, or 22.5°. In other embodiments, the X-ray powder diffraction pattern of Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,1-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) includes an XRP diffraction peak at two-theta angles of approximately 5.7 and 15.3°. In some embodiments, the X-ray powder diffraction pattern of Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,1-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) includes an XRP diffraction peak at two-theta angles of approximately 5.7°. In certain embodiments, crystalline Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) has an X-ray diffraction pattern substantially as shown in FIG. **5**.

In some embodiments, crystalline Form II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 5.7 and 15.3°. In certain embodiments, crystalline Form II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 5.7°, approximately 15.3°, and approximately 22.5°. In some embodiments, crystalline Form II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 5.7°, approximately 14.2°, approximately 15.3°, and approximately 22.5°. In other embodiments, crystalline Form II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 5.7°, approximately 14.2°, approximately 15.3°, approximately 15.9°, and approximately 22.5°. In yet other embodiments, crystalline Form II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 5.7°, approximately 14.2°, approximately 15.3°, approximately 15.9°, approximately 18.6°, and approximately 22.5°.

In various embodiments, crystalline Form II has an endothermic differential scanning calorimetric (DSC) thermogram. In some embodiments, crystalline Form II has a DSC thermogram comprising an endothermic event with an onset temperature of about 143° C. and a peak at about 155° C.; and another endothermic event with an onset temperature of about 232° C. and a peak at about 235° C.

In yet another embodiment, crystalline Form II has a DSC thermogram substantially as shown in FIG. **6**. In yet another embodiment, crystalline Form II has a thermal gravimetric analysis (TGA) plot comprising a mass loss of about 2.2% when heated from about 25° C. to about 140° C. In still another embodiment, crystalline Form II has a TGA plot substantially as shown in FIG. **6**.

In various embodiments, crystalline Form II has a gravimetric vapor system (GVS) plot. In some embodiments, crystalline Form II exhibit a mass increase of about 0.5% when subjected to a an increase in relative humidity from about 0% to about 95% relative humidity. In certain embodiments mass gained upon adsorption is lost when the relative humidity (RH) is decreased back to about 0% RH. In yet another embodiment, crystalline Form II exhibit a gravimetric vapor system plot substantially as shown in FIG. **7**. In certain embodiments, Form II is substantially non-hygroscopic. In certain embodiments, the XRPD pattern of Form II material is substantially unchanged following the adsorption/desorption analysis. In certain embodiments, Form II is stable with respect to humidity. In still another embodiment, crystalline Form II has aqueous solubility of about 18.5 mg/mL at pH 5.1.

In certain embodiments Form II may be characterized by particle analysis. In certain embodiments, a sample of Form II comprises particles having birefringent lath shaped morphology. In yet another embodiment, a sample of Form II comprises particles of about 100, about 90, about 80, about 70, about 60, about 50, about 40, about 30, about 20, about 10, about 5 μM in length. In some embodiments, a sample of Form II comprises particles of about 100, about 70, about 60, about 40, about 20, about 10 μM in length. In yet another embodiment, a sample of Form II comprises particles of about 100 μM in length.

In certain embodiments, crystalline form of Formula I in Form II may contain no less than about 95%, no less than about 97%, no less than about 98%, no less than about 99%, or no less than about 99.5% by weight of the salt of Formula I. The crystalline form may also contain no less than about 90%, no less than about 95%, no less than about 98%, no less than about 99%, or no less than 99.5% by weight of crystal Form II.

Valbenazine Ditosylate Form III

In another embodiment, is a crystalline form of (S)-(2R, 3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) or an isotopic variant thereof or solvate thereof; wherein the crystalline form is Form III.

In various embodiments, crystalline Form III of (S)-(2R, 3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate di(4-methylbenzenesulfonate) (Formula I) has an X-ray diffraction pattern. In some embodiments, the X-ray diffraction pattern of Form III of (S)-(2R,3R,11bR)-

In various embodiments, crystalline Form I of Formula II has an endothermic differential scanning calorimetric (DSC) thermogram. In some embodiments, crystalline Form I has a DSC thermogram comprising an endothermic event with onset temperature of about 240° C. and a peak at about 250° C.

In yet another embodiment, crystalline Form I of Formula II has a DSC thermogram substantially as shown in FIG. **21**. In yet another embodiment, crystalline Form I of Formula II has a thermal gravimetric analysis (TGA) plot substantially as shown in FIG. **21**.

In various embodiments, crystalline Form I of Formula II has a gravimetric vapor system (GVS) plot. In some embodiments, crystalline Form I exhibit a mass increase of about 14% when subjected to a an increase in relative humidity from about 0% to about 90% relative humidity. In yet another embodiment, crystalline Form I of Formula II exhibit a gravimetric vapour system plot substantially as shown in FIG. **22**. In certain embodiments, the XRPD pattern of Form I of Formula II is substantially changed following the adsorption/desorption analysis. In another embodiment, Form I of Formula II converts to Form II upon storage at about 25° C. and about 92% relative humidity for about 7 days. In another embodiment, Form I of Formula II converts to Form II upon storage at about 40° C. and about 75% relative humidity for about 7 days. In still another embodiment, crystalline Form I of Formula II has aqueous solubility above 90 mg/mL at pH 4.1.

In certain embodiments Form I of Formula II may be characterized by particle analysis. In certain embodiments, a sample of Form II comprises particles having birefringent lath shaped morphology. In yet another embodiment, a sample of Form I of Formula II comprises particles of about 100, about 90, about 80, about 70, about 60, about 50, about 40, about 30, about 20, about 10, about 5 μM in length. In some embodiments, a sample of Form I of Formula II comprises particles of about 100, about 70, about 60, about 40, about 20, about 10 μM in length. In yet another embodiment, a sample of Form I of Formula II comprises particles of about 150 μM in length.

In certain embodiments, crystalline form of Formula II in Form I may contain no less than about 95%, no less than about 97%, no less than about 98%, no less than about 99%, or no less than about 99.5% by weight of the salt of Formula II. The crystalline form may also contain no less than about 90%, no less than about 95%, no less than about 98%, no less than about 99%, or no less than 99.5% by weight of crystal Form I.

Valbenazine Dihydrochloride Form II

In another embodiment, is a crystalline form of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate dihydrochloride (Formula II) or an isotopic variant thereof or solvate thereof; wherein the crystalline form is Form II.

In various embodiments, crystalline Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate dihydrochloride (Formula II) has an X-ray diffraction pattern. In some embodiments, the X-ray diffraction pattern of Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate dihydrochloride (Formula II) includes an XRP diffraction peak at two-theta angles of approximately 4.8, 13.3, and 24.9°. In some embodiments, the X-ray powder diffraction pattern of Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate dihydrochloride (Formula II) includes an XRP diffraction peak at two-theta angles of approximately 4.8, 13.3 or 24.9°. In certain embodiments, the X-ray powder diffraction pattern of Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,1-a]isoquinolin-2-yl 2-amino-3-methylbutanoate dihydrochloride (Formula II) includes an XRP diffraction peak at two-theta angles of approximately 4.8°. In certain embodiments, crystalline Form II of (S)-(2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-2,3,4,6,7,11b-hexahydro-1H-pyrido[2,l-a]isoquinolin-2-yl 2-amino-3-methylbutanoate dihydrochloride (Formula II) has an X-ray diffraction pattern substantially as shown in FIG. **23**.

In some embodiments, crystalline Form II of Formula II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 4.8° and approximately 24.9°. In certain embodiments, crystalline Form II of Formula II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 4.8°, approximately 13.3°, and approximately 24.9°. In some embodiments, crystalline Form II of Formula II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 4.8°, approximately 13.3°, approximately 14.1°, and approximately 24.9°. In yet other embodiments, crystalline Form II of Formula II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 4.3°, approximately 4.8°, approximately 13.3°, approximately 14.1°, and approximately 24.9°. In some embodiments, crystalline Form II of Formula II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 4.3°, approximately 4.8°, approximately 13.3°, approximately 14.1°, approximately 18.4°, and approximately 24.9°. In other embodiments, crystalline Form II of Formula II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 4.3°, approximately 4.8°, approximately 8.7°, approximately 13.3°, approximately 14.1°, approximately 18.4°, and approximately 24.9°. In other embodiments, crystalline Form II of Formula II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 4.3°, approximately 4.8°, approximately 8.4°, approximately 8.7°, approximately 13.3°, approximately 14.1°, approximately 18.4°, and approximately 24.9°. In yet other embodiments, crystalline Form II of Formula II has one or more characteristic XRP diffraction peaks at two-theta angles of approximately 4.3°, approximately 4.8°, approximately 8.4°, approximately 8.7°, approximately 13.3°, approximately 14.1°, approximately 14.6°, approximately 18.4°, and approximately 24.9°.

In various embodiments, crystalline Form II of Formula II has an endothermic differential scanning calorimetric (DSC) thermogram. In some embodiments, crystalline Form II has a DSC thermogram comprising an endothermic event with onset temperature of about 80° C. and a peak at about 106° C.

In yet another embodiment, crystalline Form II of Formula II has a DSC thermogram substantially as shown in FIG. **24**. In yet another embodiment, crystalline Form II of Formula II has a thermal gravimetric analysis (TGA) plot comprising a mass loss of about 10% when heated from about 25° C. to about 100° C. In still another embodiment, crystalline Form II of Formula II has a TGA plot substantially as shown in FIG. **24**.

In various embodiments, crystalline Form II of Formula II has a gravimetric vapor system (GVS) plot. In some embodi-

TABLE 19-continued

Summary of salts formed.											
Synthetic details			Characterization			Aqueous solubility		Stability to humidity			
			Eq		mp onset	mg/mL free		40° C./75%	25° C./92%	GVS	
Salt form	Yield	Purity	counterion	XRPD	(° C.)	form	pH	RH	RH	uptake	
besylate	86%	99.1%	2.0 by NMR	crystalline	239	25.1	4.74	unchanged	unchanged	3.7%	
hydro-bromide	61%	98.7%	2.0 by IC	crystalline	158, 248	85.3	3.38	form change	form change	>3.9%	
hydro-chloride	n/a	99.0%	2.1 by IC	crystalline	244	>90	4.09	form change	form change	15.2% (form change)	

All the salts formed showed bis stoichiometry and purity comparable to the input material.

The examples set forth above are provided to give those of ordinary skill in the art with a complete disclosure and description of how to make and use the embodiments, and are not intended to limit the scope of the disclosure. Modifications of the above-described modes for carrying out the disclosure that are obvious to persons of skill in the art are intended to be within the scope of the following claims. All publications, patents, and patent applications cited in this specification are incorporated herein by reference as if each such publication, patent, or patent application were specifically and individually indicated to be incorporated herein by reference.

What is claimed is:

1. A crystalline form of (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester tosylate salt having a differential scanning calorimetric (DSC) peak temperature within 2% of 243° C.

2. The crystalline form of claim **1**, wherein the DSC peak temperature is within 1% of 243° C.

3. The crystalline form of claim **1**, wherein the DSC peak temperature is within 0.5% of 243° C.

4. The crystalline form of claim **1**, wherein the crystalline form has an X-ray powder diffraction (XRPD) pattern comprising a peak at a two-theta angle of 6.3°±0.2°.

5. The crystalline form of claim **1**, wherein the crystalline form has an X-ray powder diffraction (XRPD) pattern comprising a peak at a two-theta angle of 17.9°±0.2°.

6. The crystalline form of claim **1**, wherein the crystalline form has an X-ray powder diffraction (XRPD) pattern comprising a peak at a two-theta angle of 19.7°±0.2°.

7. The crystalline form of claim **1**, wherein the crystalline form is stable upon exposure to about 25° C. and about 60% relative humidity.

8. The crystalline form of claim **1**, wherein the crystalline form has a D90 particle size of about 70 μM in length.

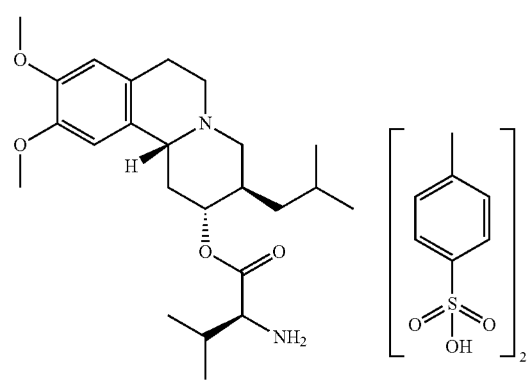
9. The crystalline form of claim **1**, wherein the crystalline form has a D10 particle size of about 10 μM in length.

10. The crystalline form of claim **1**, wherein the crystalline form has a purity of no less than 97% by weight of (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester tosylate salt.

11. The crystalline form of claim **1**, wherein the crystalline form has a purity of no less than 98% by weight of (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester tosylate salt.

12. The crystalline form of claim **1**, wherein the crystalline form has a purity of no less than 97% by weight of (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,l-a]isoquinolin-2-yl ester tosylate salt; and has an X-ray powder diffraction (XRPD) pattern comprising peaks at two-theta angles of 6.3°±0.2°, 17.9°±0.2°, and 19.7°±0.2°.

13. The crystalline form of claim **1**, wherein the (S)-2-amino-3-methyl-butyric acid (2R,3R,11bR)-3-isobutyl-9,10-dimethoxy-1,3,4,6,7,11b-hexahydro-2H-pyrido[2,1-a]isoquinolin-2-yl ester tosylate salt is:



14. A pharmaceutical composition comprising the crystalline form of claim **1** and a pharmaceutically acceptable carrier.

15. The pharmaceutical composition of claim **14**, wherein the composition is formulated for oral administration.

16. The pharmaceutical composition of claim **14**, wherein the pharmaceutical composition is a unit dosage form.

17. The pharmaceutical composition of claim **14**, wherein the pharmaceutical composition is in the form of a capsule.

18. A pharmaceutical composition comprising the crystalline form of claim **12** and a pharmaceutically acceptable carrier.

19. The pharmaceutical composition of claim **18**, wherein the composition is formulated for oral administration.